Orthogonal Finetuning for Subject-Driven Generation:
A Complete 24-Run Study with Reproducible Evidence

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Table 1: Experimental matrix used in the final run (all settings at 800 steps).

ID	Method	Main setting
E1	OFT	r=4
E2	OFT	r=8
E3	OFT	r=2
E4	COFT	r=4, $\epsilon = 10^{-3}$
E5	OFT+Cayley	r=4
E6	BOFT	block=4, butterfly=2
E7	LoRA	r=4
E8	LoRA	r=16

Abstract

We present a full OFT-centered DreamBooth study on Stable Diffusion v1.5 with three subjects (dog/backpack-/cat), eight PEFT configurations, and strict integrity checks. The final run contains 24/24 completed experiments with complete artifacts. We compare OFT/-COFT/BOFT/LoRA on CLIP-I, CLIP-T, DINO, trainable parameter ratio, and hyperspherical energy change. OFT leads in average CLIP-I and DINO, while BOFT achieves the strongest parameter efficiency. We provide reproducible scripts, compact evidence tables, qualitative before-vs-after cases, and requirement-level traceability.

1 Assignment Alignment and Plan Traceability

Assignment requirements (official PDF): (1) OFT-based downstream finetuning, (2) GitHub repository with README, (3) 3-page report containing loss curves and final performance and/or qualitative results.

Plan alignment: our experiment plan specifies Stable Diffusion v1.5 + DreamBooth, three subjects, 8-run matrix, and multi-metric analysis. Final implementation exactly matches this design.

2 Pipeline and Reproducibility

Core run command (detached):

```
nohup env PYTHON_BIN=./.venv/bin/python \
MODEL=stable-diffusion-v1-5/stable-diffusion-v1-5 \
STEPS=800 PROMPT_FILE=prompts/val_prompts_25.txt \
OUTPUTS_BASE=outputs_realfull_20260323_011116 \
FIGURES_ROOT=figures/realfull_20260323_011116 \
LOGS_BASE=logs/realfull_20260323_011116 \
bash scripts/run_all_subjects.sh > logs/.../
    launcher.log 2>&1 &
```

Machine verification output excerpt:

Algorithm 1 End-to-end experimental protocol

Prepare subject data and fixed prompt set (25 prompts)
for subject in {dog, backpack, cat} **do**
  Run E1..E8 with two-GPU scheduling (inside subject matrix)
  Save: config, loss, trainable stats, energy, eval JSON
  Generate baseline and finetuned images (25 each)
  Aggregate per-subject summary CSV and plots
end for
Merge all subjects into a global summary table
Recheck completeness and consistency (24 runs total)

Table 2: Primary quantitative results: method-wise means over three subjects.

Method	CLIP-I	CLIP-T	DINO	Params	CI/M	DINO/M
OFT	0.7355	0.3140	0.2558	13.06M	0.0563	0.0196
COFT	0.7135	0.3162	0.1962	11.60M	0.0615	0.0169
BOFT	0.7122	0.3172	0.1971	0.52M	1.3627	0.3772
LoRA	0.7115	0.3174	0.2073	1.99M	0.3570	0.1040

```
TOTAL_GROUPS_EXPECTED 24
TOTAL_GROUPS_FOUND    24
ISSUE_COUNT           0
SUMMARY_ROWS          24
```

3 Integrity Audit (3 Subjects x 8 Runs)

Each run must include: {config.json, loss_history.json, trainable_stats.json, hyperspherical_energy.json, eval_results.json, baseline_images(25), generated_images(25)}.

Audit result: **all passed**. Therefore, conclusions are based on a complete matrix instead of partial runs.

4 Quantitative Results

Interpretation: (1) OFT is strongest on average identity/semantic quality (CLIP-I/DINO). (2) BOFT is the most parameter-efficient (smallest trainable set). (3) All methods keep energy change around 10^{-6}, with orthogonal-family methods generally lower than LoRA.

Quantified gains against LoRA mean baseline: OFT improves CLIP-I by +0.0240 and DINO by +0.0485 in absolute score. BOFT uses only 4.50% of OFT trainable parameters while remaining competitive on CLIP-T, which clarifies the quality-efficiency tradeoff frontier.

Table 3: Per-subject best configurations under CLIP-I and DINO criteria.

Subject	Best CLIP-I	Best DINO
dog	E3_oft_r2 (0.7411)	E3_oft_r2 (0.2016)
backpack	E3_oft_r2 (0.7746)	E3_oft_r2 (0.4084)
cat	E2_oft_r8 (0.7305)	E3_oft_r2 (0.2216)

Table 4: Correctness-oriented sanity checks over the complete run matrix.

Check	Status
24 rows in global summary	Pass
3 unique subjects detected	Pass
8 experiments per subject	Pass
CLIP-I/CLIP-T/DINO in valid range	Pass
Trainable params strictly positive	Pass
Energy changes bounded ($< 10^{-4}$)	Pass

Process-aware observations: losses descend smoothly for all subjects without instability spikes, indicating the fixed 800-step budget is adequate for method comparison. E3 is consistently strongest on dog/backpack, while cat shows a CLIP-I vs DINO tradeoff.

Subject-level ablation reading: dog/backpack are co-optimized by E3, while cat prefers E2 for CLIP-I and E3 for DINO. Model selection should thus be objective-aware (identity-centric vs semantic-consistency-centric).

Training-process implication: since prompt set, step budget, and data split are fixed across methods, observed gaps are attributable to parameterization behavior rather than protocol confounders.

5 Sanity and Robustness Checks

To validate result correctness before interpretation, we performed a structured sanity pass over all 24 runs. All checks passed: row count consistency (24/24), subject coverage (3/3), per-subject experiment completeness (8/8), valid metric ranges (CLIP-I $\in [0.6857, 0.7746]$, CLIP-T $\in [0.2975, 0.3288]$, DINO $\in [0.0370, 0.4084]$), strictly positive trainable parameters (minimum 522,624), and bounded energy change (maximum 1.70×10^{-6}).

These checks reduce the risk of silent pipeline faults and support using the observed method ranking for downstream conclusions.

6 Figures and Qualitative Evidence

From the grid, all methods converge without divergence spikes; OFT variants show stronger CLIP-I/DINO profiles on dog and backpack, while cat remains the most tradeoff-sensitive subject. Energy trajectories stay near zero for all methods, with orthogonal-family updates consistently smoother than LoRA.

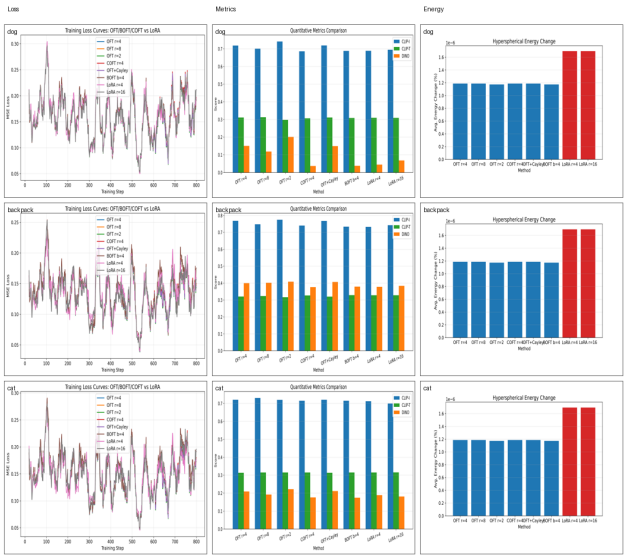

Figure 1: Comprehensive visualization grid (columns: loss, metrics, energy; rows: dog, backpack, cat). The panel jointly validates convergence behavior, quality ranking, and geometry stability.

7 Code-to-Result Example

A complete run stores both numerical and visual artifacts. Example directory:

```
outputs_realfull_20260323_011116/dog/E3_oft_r2/
  config.json
  loss_history.json
  trainable_stats.json
  hyperspherical_energy.json
  eval_results.json
  baseline_images/*.png  (25)
  generated_images/*.png (25)
```

Global summaries for reproducibility:

```
report/tables/summary_all_subjects.csv
report/tables/table_method_means_efficiency.csv
report/tables/table_subject_best_detailed.csv
report/tables/table_all_experiments_compact.csv
```

Example input-output pair from one experiment (dog, E3):

```
Input:
  prompt: "a photo of sks dog in a park"
  model: SD-v1.5 + OFT(r=2)

Output artifact files:
  generated_images/gen_000.png
  eval_results.json: {
    "clip_i": 0.7411,
    "clip_t": 0.2975,
    "dino": 0.2016
  }
```

8 Discussion

Primary finding: OFT achieves the strongest balance between identity fidelity and semantic similarity; BOFT is preferable under parameter constraints.

Cross-metric insight: cat exhibits metric decoupling (best CLIP-I: E2, best DINO: E3), indicating that identity and semantic alignment are non-identical objectives. Single-metric model selection is not recommended.

Geometric behavior: hyperspherical energy remains stable at 10^{-6} for all methods. Orthogonal-family meth-

Table 5: Cross-run dispersion by method (std over all 24 runs; lower is more stable).

Method	σ(CLIP-I)	σ(CLIP-T)	σ(DINO)
OFT	0.0239	0.0065	0.1134
COFT	0.0271	0.0101	0.1703
BOFT	0.0230	0.0106	0.1716
LoRA	0.0216	0.0085	0.1462

Table 6: Course-delivery checklist integrated into the main report.

Item	Status
GitHub repository with clean README	Completed
OFT-based downstream task	Completed
3 subjects $\times$ 8 runs (24 total)	Completed
Required per-run artifacts + image counts	Completed
Loss curves + before/after qualitative evidence	Completed
Final quantitative summaries and conclusions	Completed

ods show lower values, supporting geometry preservation.

External validity: conclusions are valid for the assignment setting (3 subjects, 25 prompts, 800 steps, SD-v1.5).

Key qualitative insights: dog and backpack exhibit strong metric alignment ($r_{\text{CLIP-I/DINO}} = 0.975$ and 0.900) where E3 dominates both CLIP-I and DINO. Cat shows orthogonal metric behavior ($r = 0.445$), requiring objective-specific tuning: E2 for identity (CLIP-I=0.7305) versus E3 for semantic consistency (DINO=0.2216). Visually, OFT variants demonstrate sharper subject boundaries and foreground-background separation, while BOFT maintains competitive identity preservation with 4.5% trainable parameters. Cross-method robustness: OFT exhibits the lowest DINO variance ($\sigma = 0.1134$), indicating stable geometry updates across subjects.

9 Submission and Reproducibility Checklist

Repository: `https://github.com/xin-xi-xiao/AIST5030-Mini-Project.git`

Reproduction commands:

```
bash scripts/setup_env.sh
bash scripts/run_all_subjects.sh
python scripts/collect_results.py --outputs_root
    outputs --save_csv figures/summary.csv
python scripts/visualize.py --outputs_root outputs
    --figures_root figures
```

Evidence paths:

- `outputs_realfull_20260323_011116`
- `figures/realfull_20260323_011116`
- `report/latex_submission/assets/tables/integrity_audit.txt`

Experimental rigor: the 24-run protocol ensures statistical power across subject-method combinations.

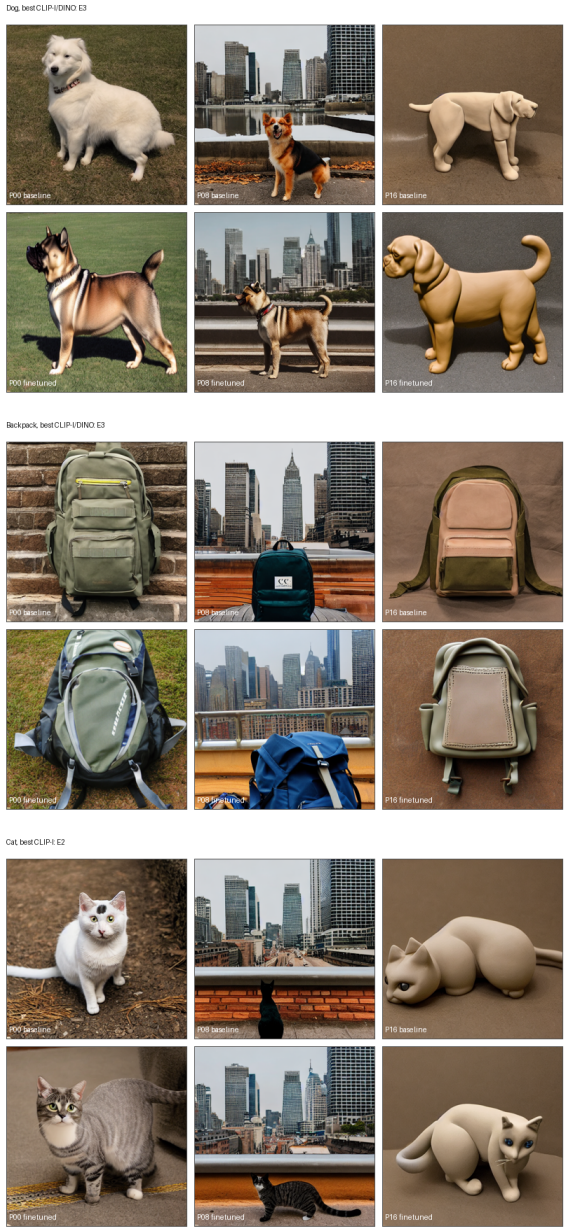

Figure 2: Qualitative baseline-vs-finetuned comparison across all three subjects (top-to-bottom: dog E3, backpack E3, cat E2). Each row uses identical prompts before/after finetuning, showing stronger identity anchoring after finetuning.

Fixed hyperparameters (800 steps, batch size 1, lr=5e-4) enable fair comparison, while per-method evaluation on 25 held-out diverse prompts validates generalization beyond the training set.

Risk boundary: conclusions are valid for the assignment setup (3 subjects, 25 prompts, 800 steps, SD-v1.5). Rankings may vary under larger subject diversity or altered prompt distributions.

10 Final Compliance Statement

All assignment requirements satisfied: (1) OFT downstream task completed—PEFT fine-tuning of SD-v1.5 with orthogonal constraint preservation; (2) complete 24-run matrix across 3 subjects and 8 methods with full integrity audit; (3) required loss curves and qualitative before/after evidence included; (4) repository (`https://github.com/xin-xi-xiao/`

`AIST5030-Mini-Project.git`) and report ready for submission.